Introduction to Transport Networks

Network Infrastructures
MSc. Cybersecurity

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Why do we need transport networks?

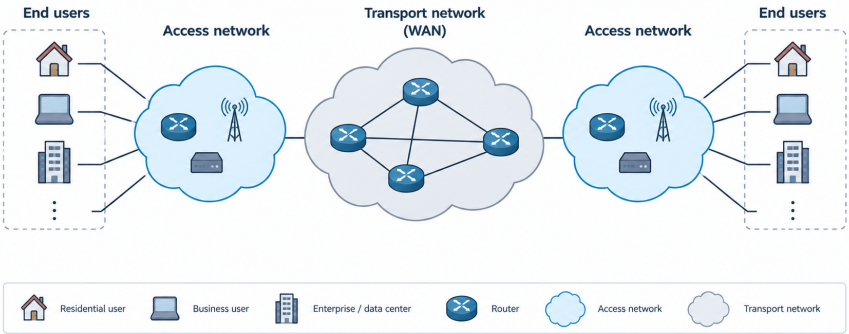
- Access networks provide connectivity between end users and a network operator.
- Access networks typically cover a relatively limited geographical area.
- However, users need to communicate with destinations located beyond their local access network.
- This requires an infrastructure capable of interconnecting geographically separated networks.

Transport Networks

Transport networks provide connectivity between distant network regions, carrying traffic aggregated by access networks.

From access to transport

1 What is a Transport Networks?





Transport networks and WANs

1 What is a Transport Networks?

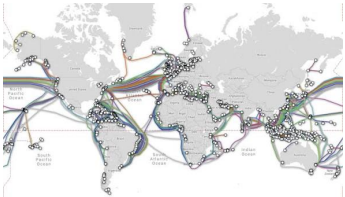
- A transport network may span cities, regions, countries or continents.
- It interconnects access regions and carries already aggregated traffic.
- The access/transport boundary depends on the operator and technology.

WAN

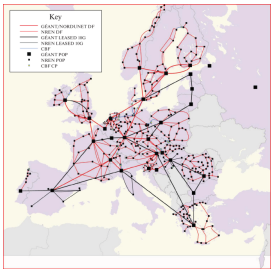
Wide Area Network emphasizes the large geographical scope of this infrastructure.

Examples of Transport Networks

Submarine cables



GÉANT Network



Google Network



- Transport infrastructures interconnect geographically distributed network locations.
- Different organizations deploy and operate their own transport infrastructures.



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Who is the transport provider's customer?

Access network

- Individuals and households
- Companies and organizations
- Many customers, relatively small individual flows

Transport network

- Access providers
- Other network operators
- Large organizations
- Fewer customers, large traffic aggregates



Customer–provider relationships are recursive

End user $\longrightarrow$ Access provider

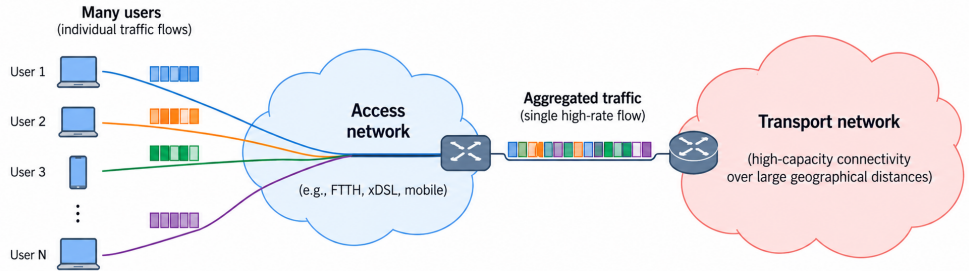
Access provider $\longrightarrow$ Transport carrier

Transport carrier $\longrightarrow$ Other networks

The client of a transport service can be an entire network or a router, not necessarily a person.

Traffic aggregation

2 What Makes Transport Networks Different?



1. Many individual flows
Each user generates a low-rate flow (e.g., web browsing, video, VoIP, ...).

2. Traffic aggregation
The access network aggregates many individual flows into a single, much larger flow towards the transport network.

3. High-rate transport
The transport network carries the aggregated traffic over long distances with high-capacity links (e.g., optical fiber).



Tributary flows and aggregate capacity

Conceptual aggregation

$$F_{\text{agg}} = \sum_{i=1}^{N} F_i$$

- Individual access flows are combined upstream.
- Transport links handle much larger traffic volumes than single access links.
- Real traffic is bursty: subscribed peak rates are not continuously generated.



Example: 1,000 FTTH users

- Each user subscribes to a 1 Gbps access service.
- Sum of nominal access rates:

$$1000 \times 1\ \text{Gbps} = 1\ \text{Tbps}$$

- If each generates an average of 50 Mbps in a busy period:

$$1000 \times 50\ \text{Mbps} = 50\ \text{Gbps}$$

Statistical multiplexing

Upstream capacity need not equal the sum of all subscribed peak rates.



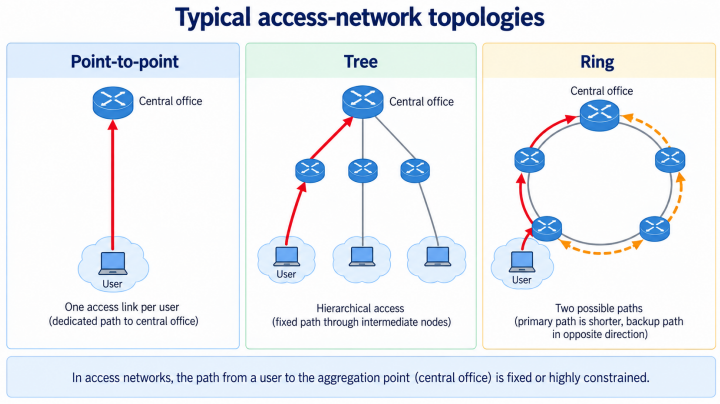
Transport Networks: High Capacity

- Access networks aggregate traffic generated by multiple end users.
- Transport networks further aggregate traffic originating from multiple access networks.
- Consequently, transport links and switching nodes must support very high data rates, significantly exceeding those of individual access connections.

Key characteristic

High capacity is a fundamental requirement of transport networks.

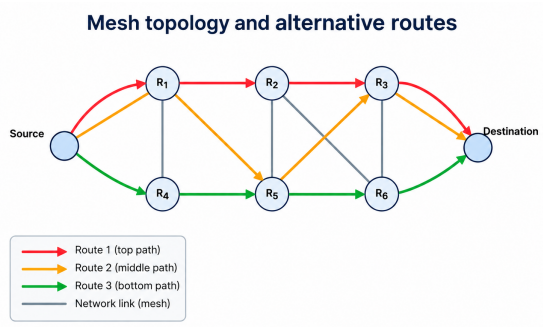
Typical access topologies



Point-to-point and tree structures typically determine a unique path; rings can provide protection alternatives.

Transport networks: mesh topologies

2 What Makes Transport Networks Different?





Routing becomes a decision problem

- A mesh can offer multiple paths between the same endpoints.
- The operator must choose which intermediate nodes to traverse.
- Path choice affects resource utilization, resilience, delay and available capacity.

Example

Using only one route can overload some links while leaving alternative paths underutilized.



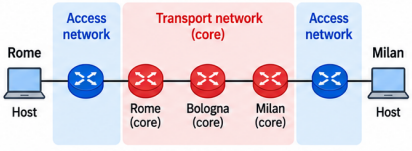
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An end-to-end example: Rome to Milan

3 Physical Transport Infrastructure



(a) End-to-end path between Rome and Milan

```
$ traceroute milan.example.net
traceroute to milan.example.net (203.0.113.10), 30 hops max, 60 byte packets
 1   gw.rome.example.net              0.4 ms   0.5 ms   0.4 ms   Access network (Rome)
 2   core1.rome.example.net           2.1 ms   2.3 ms   2.0 ms
 3   r1.roma-bologna.example.net      7.8 ms   7.5 ms   7.6 ms   Transport network (Rome → Milan)
 4   r2.bologna.example.net          12.1 ms  11.8 ms  12.0 ms
 5   r3.bologna-milan.example.net    16.4 ms  16.1 ms  16.3 ms
 6   core1.milan.example.net         21.7 ms  21.9 ms  21.6 ms
 7   gw.milan.example.net            26.8 ms  26.5 ms  26.7 ms   Access network (Milan)
 8   milan.example.net (203.0.113.10) 28.9 ms  28.7 ms  28.8 ms
```

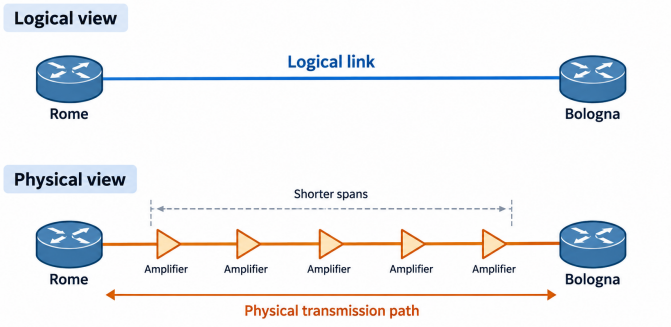
(b) Example traceroute from Rome to Milan

Traceroute exposes Layer-3 hops, not necessarily the underlying physical route or exact router locations.

One logical link, multiple physical spans

3 Physical Transport Infrastructure

- Two remote routers can appear directly connected at Layer 3.
- Their logical link does not describe the actual physical path.
- Long-distance signals experience propagation impairments.
- Intermediate infrastructure, including amplification, may be necessary.

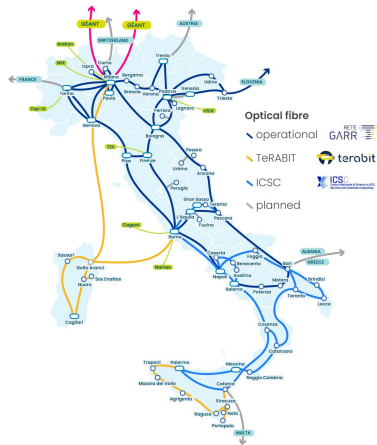


Why not dedicate a physical path to each client?

3 Physical Transport Infrastructure

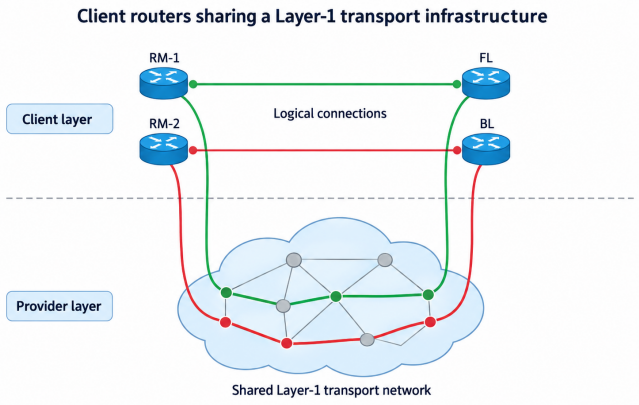
- Nationwide physical infrastructure is costly to deploy and operate.
- Dedicated end-to-end facilities may leave capacity unused.
- Multiple clients can share common physical resources.

- **Architectural requirement:** Provide separate logical connections over a shared physical infrastructure.



A shared Layer-1 transport provider

3 Physical Transport Infrastructure





Client and provider views

3 Physical Transport Infrastructure

Client layer (e.g., IP)

- Requests connectivity
- Sees logical links
- Does not need internal physical details

Transport provider (Layer 1)

- Owns physical resources
- Maps client connections onto paths
- Shares infrastructure among clients



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4 Summary and next steps

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Key architectural principles

4 Summary and next steps

1. Transport interconnects geographically separated network regions.
2. Its customers are often other networks.
3. It carries large traffic aggregates.
4. Mesh topology makes routing central.
5. Logical connectivity is abstracted from physical realization.
6. Lower-layer infrastructure is shared across clients.

Coming next

4 Summary and next steps

- We will follow a bottom-up approach through the transport network stack.

